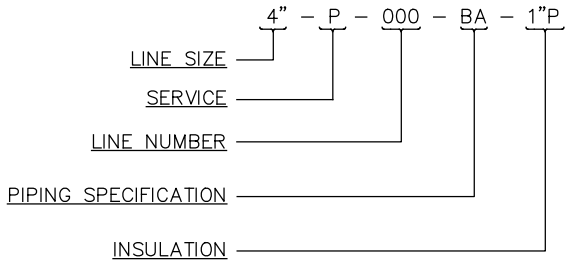
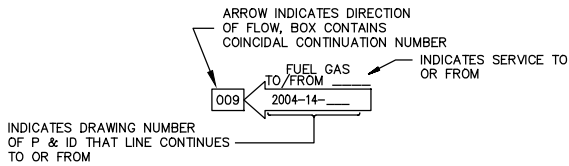


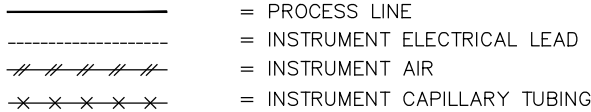
LINE DESIGNATION



LINE CONTINUATION ARROW



LINE SYMBOLS



MATERIAL SPECIFICATIONS DESIGNATIONS
(IN ACCORDANCE WITH ASME/ANSI B31.3 STANDARD)

- A1 – CLASS 150 RF, CARBON STEEL
1 1/2" & SMLR SCH 80 SEAMLESS ASTM A106 Gr B
2"-24" STANDARD WEIGHT SEAMLESS ASTM A53 Gr B
26" & UP CALCULATE WALL THICKNESS
- A3 – CLASS 300 RF, CARBON STEEL
1 1/2" & SMLR SCH 80 SEAMLESS ASTM A106 Gr B
(EXCEPT SCH 160 SHALL BE USED FOR 1/2" & 3/4")
2"-12" STANDARD WEIGHT SEAMLESS ASTM A53 Gr B
14"-24" CALCULATE WALL THICKNESS (0.375" MIN.)
SEAMLESS ASTM A53 GR B
26" & UP CALCULATE WALL THICKNESS
- A6 – CLASS 600 RF, CARBON STEEL
1 1/2" & SMLR SCH 160 SEAMLESS ASTM A106 Gr B
2"-12" SCH 80 SEAMLESS ASTM A53 Gr B
14"-24" CALCULATE WALL THICKNESS (0.375" MIN.)
SEAMLESS ASTM A53 Gr B
- B1 – CLASS 150 RF, 1 1/4 Cr – 1/2 Mo
1 1/2" & SMLR SCH 80 SEAMLESS ASTM A335 Gr P11
2"-14" STD. WEIGHT SEAMLESS ASTM A335 Gr P11
16"-24" CALCULATE WALL THICKNESS (0.375" MIN.)
SEAMLESS ASTM A335 Gr P11
- B3 – CLASS 300 RF, 1 1/4 Cr – 1/2 Mo
1 1/2" & SMLR SCH 80 SEAMLESS ASTM A335 Gr P11
2"-14" STD. WEIGHT SEAMLESS ASTM A335 Gr P11
16"-24" CALCULATE WALL THICKNESS (0.375" MIN.)
SEAMLESS ASTM A335 Gr P11
- B6 – CLASS 600 RF, 1 1/4Cr-1/2 Mo
1 1/2" & SMLR SCH 160 SEAMLESS ASTM A335 Gr P11
2"-12" SCH 80 SEAMLESS ASTM A335 Gr P11
14"-24" CALCULATE WALL THICKNESS (0.375" MIN.)
ASTM A335 Gr P11

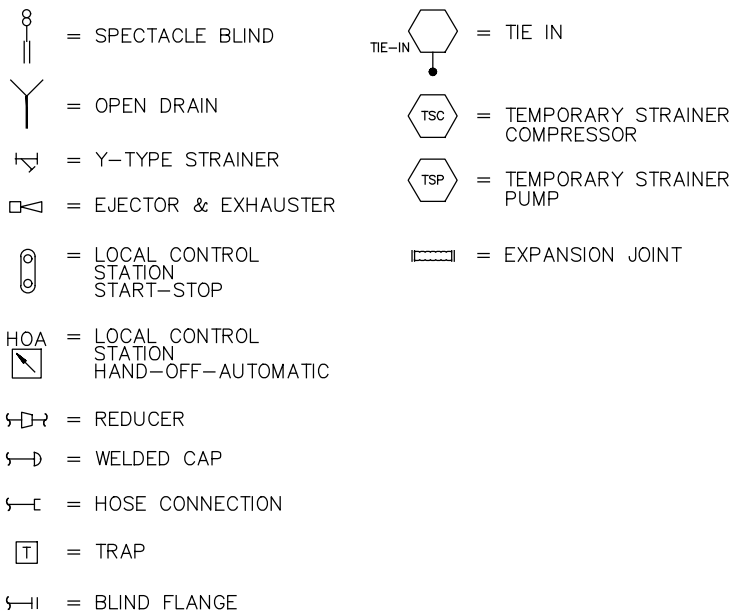
SERVICE ABBREVIATIONS

- FG – FUEL GAS
BD – BLOWDOWN
IA – INSTRUMENT AIR
P – PROCESS
HH – HYDROGEN
S – STEAM
SW – SOUR WATER
CWS– COOLING WATER SUPPLY
CWR– COOLING WATER RETURN
N – NITROGEN
KM – CHEMICAL INJECTION
SC– STEAM CONDENSATE
SE– STEAM EXHAUST
H2S– HYDROGEN SULFIDE

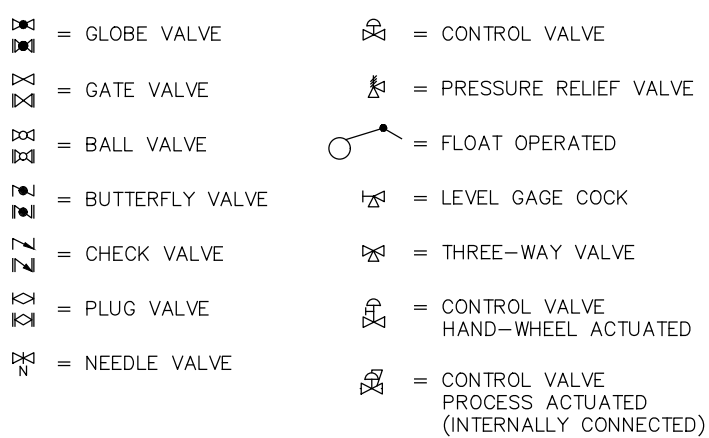
INSULATION ABBREVIATIONS

- P – PERSONNEL PROTECTION
H – HEAT CONSERVATION

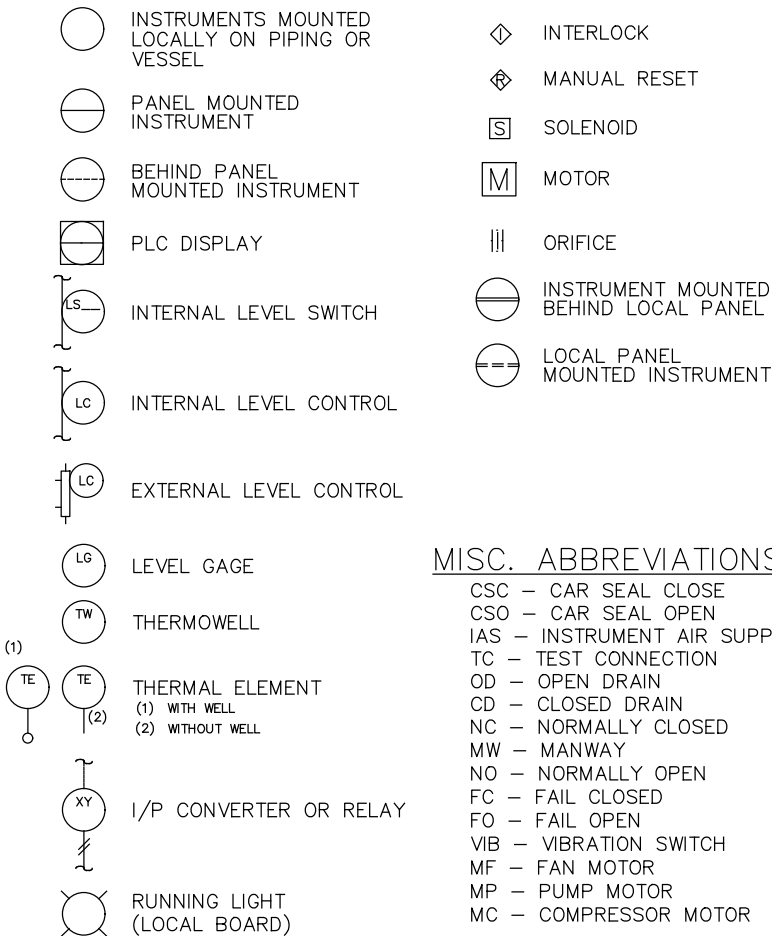
MISCELLANEOUS SYMBOLS



VALVE SYMBOLS



INSTRUMENT SYMBOLS



MISC. ABBREVIATIONS

- CSC – CAR SEAL CLOSE
CSO – CAR SEAL OPEN
IAS – INSTRUMENT AIR SUPPLY
TC – TEST CONNECTION
OD – OPEN DRAIN
CD – CLOSED DRAIN
NC – NORMALLY CLOSED
MW – MANWAY
NO – NORMALLY OPEN
FC – FAIL CLOSED
FO – FAIL OPEN
VIB – VIBRATION SWITCH
MF – FAN MOTOR
MP – PUMP MOTOR
MC – COMPRESSOR MOTOR

AP	ANALYSIS CONNECTION	TAHL	TEMPERATURE ALARM HIGH LOW
PP	PRESSURE CONNECTION	TSH	TEMPERATURE SWITCH HIGH
HIS	HAND INDICATING SWITCH	TSL	TEMPERATURE SWITCH LOW
FY	FLOW RELAY OR CONVERTER	TAH	TEMPERATURE ALARM HIGH
LY	LEVEL RELAY OR CONVERTER	TAL	TEMPERATURE ALARM LOW
PY	PRESSURE RELAY OR CONVERTER	PV	PRESSURE CONTROL VALVE
TY	TEMPERATURE RELAY OR CONVERTER	PVC	PRESSURE REGULATOR, SELF CONTAINED
FR	FLOW RECORDER	PSV	PRESSURE SAFETY RELIEF VALVE
FRC	FLOW RECORDER CONTROLLER	FV	FLOW VALVE
FSL	FLOW SWITCH LOW	HV	HANDWHEEL VALVE
FAL	FLOW ALARM LOW	PI	PRESSURE INDICATOR
FIT	FLOW TRANSMITTER	PT	PRESSURE TRANSMITTER
PR	PRESSURE RECORDER	PIC	PRESSURE INDICATOR CONTROLLER
TI	TEMPERATURE INDICATOR	FI	FLOW INDICATOR
TR	TEMPERATURE RECORDER	FE	FLOW ELEMENT
TIC	TEMPERATURE INDICATOR CONTROLLER	FIT	FLOW INDICATOR TRANSMITTER
PSL	PRESSURE SWITCH LOW	FQ	FLOW CALCULATION
PAL	PRESSURE ALARM LOW	FIR	FLOW INDICATOR RECORDER
PDSH	PRESSURE DIFF. SWITCH HIGH	FIC	FLOW INDICATOR CONTROLLER
PDAH	PRESSURE DIFF. ALARM HIGH	AT	ANALYZER TRANSMITTER
PDV	PRESSURE DIFF. VALVE	AIR	ANALYZER INDICATOR RECORDER
PAHL	PRESSURE ALARM HIGH LOW	AQ	ANALYZER CALCULATION
LSL	LEVEL SWITCH LOW	TV	TEMPERATURE CONTROL VALVE
LAL	LEVEL ALARM LOW	XSH	SHUT DOWN SWITCH
LI	LEVEL INDICATOR	XY	VARIABLE (X) RELAY OR CONVERTER
LG	LEVEL GAGE, TUBULAR GLASS	FC	FAIL CLOSED
LGR	LEVEL GAGE REFLEX	FO	FAIL OPEN
LGT	LEVEL GAGE TRANSPARENT		
LT	LEVEL TRANSMITTER		
LC	LEVEL CONTROLLER		
LIC	LEVEL INDICATOR CONTROLLER		
LV	LEVEL VALVE		
LAH	LEVEL ALARM HIGH		
LAHL	LEVEL ALARM HIGH LOW		
LAHH	LEVEL ALARM HIGH HIGH		
LSH	LEVEL SWITCH HIGH		
LSHH	LEVEL SWITCH HIGH HIGH		
LALL	LEVEL ALARM LOW LOW		



DRAWING NO.: 2004-06-150

24	08/24/16	NO REVISIONS THIS SHEET	LMM	JRB	-
23	09/30/16	NO REVISIONS THIS SHEET	DJA	DRS	-
22	01/04/16	NO REVISIONS THIS SHEET	DJA	JRB	-
21	09/30/15	NO REVISIONS THIS SHEET	HRP	JRB	-
20	03/03/15	NO REVISIONS THIS SHEET	HRP	DAM	-
19	03/15/15	NO REVISIONS THIS SHEET	HRP	TM	-
18	08/27/14	NO REVISIONS THIS SHEET	PSC	DAM	-
26	02/02/24	BORDER CHANGE /24DDDC027	HRP	-	-
25	02/04/20	MOC 18-49	LMM	MH	DLP
REV. NO.	DATE	REVISED	REV. BY	CHK. BY	APP'D



DRAWN BY	GRIMM	DATE	2004
DESIGN		DATE	
CHECK		DATE	
ENGINEER		DATE	
APPROVER		DATE	
APPROVER		DATE	
APPROVER		DATE	

PIPING & INSTRUMENTATION DIAGRAM
(NHT) NAPHTHA HYDROTREATER
KERN ENERGY
BAKERSFIELD,CA. 93307

SCALE	DWG NO.	SHT.	REV. NO.
N.T.S	3102-D 2000	1 of 13	26